Highlights

$0.60 Is All You Need: Cost-Efficient LLM Fine-Tuning for SOC Alert Classification

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- Full label compliance achieved at only $0.60 using reasoning architectures.
- Phi4-mini achieves 100% strict F1 at 8× lower cost than data scaling.
- Cost-performance Pareto frontier across 11 LLMs and 4 training strategies.
- Reasoning models eliminate label hallucination without additional training data.
- Practical deployment guidelines for resource-constrained SOC environments.

$0.60 Is All You Need: Cost-Efficient LLM Fine-Tuning for SOC Alert Classification

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Abstract

Deploying LLMs in Security Operations Centers raises a fundamental cost question: when is fine-tuning worth the investment over traditional ML or commercial APIs? We benchmark 6 fine-tuned LLMs (0.8B–9B parameters) on training cost, inference throughput, and strict label compliance using NVIDIA A100 GPUs. The cheapest model with perfect strict F1 is Phi4-mini at **$0.60** (18 minutes). Fine-tuning is 149–181× cheaper than commercial LLM APIs (GPT-4o: $556, Claude Opus: $672 for equivalent 9,851-sample evaluation), while achieving *higher* accuracy. We discover an architecture cost paradox: Qwen3.5-0.8B (smallest model) costs $2.24 due to visual encoder overhead—3.7× more than the 7B DeepSeek-R1 at $0.70. Parameter count is not a reliable cost predictor. Traditional ML baselines (SVM: $0.00, F1=90.9%) remain the most cost-efficient option when strict compliance is not required. We propose a hybrid DT→LLM cascade that handles 99.5% of alerts via Decision Tree, reducing LLM inference to 0.5% of traffic with zero quality loss. We formalize a Total Cost of Ownership model (Definition 1) showing annual costs of $6 for a 100K-alert/day SOC with perfect F1—0.003% of the commercial API alternative. Our four-dimensional cost analysis (training + inference + opportunity + API alternative) provides the first comprehensive economic framework for LLM deployment in SOC automation.

Keywords: cost efficiency, LLM fine-tuning, SOC classification, QLoRA, parameter-efficient, inference optimization

1. Introduction

An NVIDIA A100 80GB GPU costs approximately $2 per hour on cloud platforms. When a Support Vector Machine achieves 90.9% F1 for free on CPU, LLM fine-tuning must justify its cost premium with clear economic reasoning. Yet the cost landscape for LLM deployment in production SOC environments has received surprisingly little systematic analysis—despite the average data breach costing $4.45M [21] and SOC analysts processing thousands of alerts daily.

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Our companion studies examined label compliance [14], scaling laws [16], architecture effects [19], LoRA rank sensitivity [20], multi-task regularization [18], task entropy [17], and cascade routing [15]. Here we address the *economic* dimension: what does each approach actually cost, and when is each investment worthwhile?

The cost calculation extends beyond training. A production SOC must consider four dimensions:

1. **Training cost**: The one-time GPU cost to fine-tune a model.
2. **Inference cost**: The ongoing GPU cost to classify alerts in production.
3. **Opportunity cost**: The value of the 9.1 percentage points gained over SVM.
4. **API alternative cost**: The cost of using commercial APIs (GPT-4o, Claude) instead.

We provide the first full economic analysis of parameter-efficient fine-tuning for SOC alert classification, benchmarking 6 models across all four cost dimensions. Our key findings:

1. **Fine-tuning is extremely cheap**: \$0.60–\$3.72 per model, making it 149–181× cheaper than commercial APIs.
2. **Parameter count $\neq$ cost**: The 0.8B model costs 3.7× more than the 7B model due to architectural overhead.
3. **Traditional ML is hard to beat on cost**: SVM achieves 90.9% for \$0, and the marginal value of LLM's additional 9.1 points costs \$0.066/point.
4. **Hybrid cascades optimize both**: A DT→LLM cascade reduces inference cost by 193× while maintaining 100% F1.
5. **TCO is dominated by inference**: For a 100K-alert/day SOC, annual inference cost (\$987) dwarfs training cost (\$0.60), making cascade architecture essential.

The remainder of this paper is organized as follows: Section II reviews cost-aware ML and PEFT. Section III describes the experimental setup. Section IV presents cost benchmarks and the architecture paradox. Section V analyzes inference throughput and cascade architecture. Section VI presents the formal TCO model. Section VII provides deployment decision guidance. Section VIII concludes.

2. Related Work

2.1. Cost-Aware ML Deployment

Cost efficiency in ML has been studied from multiple perspectives. Zheng et al. [24] optimized training cost through spot instance scheduling. Kaplan et al. [10] established neural scaling laws relating compute to loss. Hoffmann et al. [8] (Chinchilla) optimized the compute-data ratio, showing that many models are undertrained relative to their size. FrugalGPT [4] demonstrated 98% cost reduction by routing queries across GPT-3.5/4 API tiers based on estimated difficulty. Ong et al. [13] proposed RouteLLM for cost-efficient

API routing. The Ponemon Institute [21] reported the average data breach costs $4.45M, establishing the business case for automated SOC classification. None of these works provide a detailed TCO model for domain-specific fine-tuning.

2.2. Parameter-Efficient Fine-Tuning

QLoRA [5] cuts GPU memory by 75% via 4-bit quantization, bringing 7B+ models within reach of a single consumer GPU. LoRA [9] injects trainable low-rank adapters into frozen weights, keeping <1% of parameters active. Open-weight models like LLaMA [22] have made fine-tuning accessible to small teams. Reasoning architectures—Phi-3/Phi-4 [1] and DeepSeek-R1 [7]—achieve high performance at lower cost by fundamentally changing the cost-quality tradeoff. AdaLoRA [23] adaptively allocates rank but does not consider cost-per-F1-point.

2.3. Traditional ML for Cybersecurity

Al-Rakhami et al. [2] achieved 93% accuracy with ensemble methods on UNSW-NB15 [12]. Arp et al. [3] catalogued critical pitfalls in ML-based security evaluation. Decision Tree, SVM, and Random Forest consistently achieve >90% F1 on network intrusion detection datasets. Ferrag et al. [6] surveyed LLM applications in cybersecurity. Motlagh et al. [11] benchmarked ChatGPT on 5 datasets (45–92% F1), establishing that commercial APIs underperform domain-specific models. These traditional methods require negligible training time, establishing a strong cost baseline that LLMs must justify exceeding.

Novelty gap. No prior work has (1) demonstrated that parameter count does not predict training cost due to architectural overhead (paradox), (2) quantified the break-even point between fine-tuning and commercial APIs at <1 inference run, (3) provided a formal TCO model combining training, inference, and cascade costs, or (4) benchmarked cost-per-F1-point across 6 models, traditional ML, and commercial APIs.

3. Experimental Setup

3.1. Hardware and Pricing

All training on NVIDIA A100 80GB GPUs via the Lanta HPC system (NSTDA ThaiSC). We use $2/GPU-hour as the reference price, consistent with major cloud providers (AWS p4d.24xlarge: $3.06/hr, GCP a2-highgpu-1g: $2.48/hr, Lambda: $1.29/hr). Our $2/hr is a moderate estimate; actual costs may vary by ±50%.

3.2. Models

We evaluate 6 LLMs with QLoRA (rank 64, 4-bit NF4, bf16, 3 epochs, seed 42):

Table 1: Training profiles for 7 LLMs: GPU time, VRAM usage, cost per run, and cost-per-F1-point efficiency

Model	Params	LoRA	VRAM	Arch
Qwen3.5-0.8B	0.8B	33.6M	12GB	Multimodal
SmolLM2	1.7B	33.6M	14GB	Base
Phi4-mini	3.8B	33.6M	18GB	Reasoning
Mistral-7B	7B	33.6M	24GB	Base
DeepSeek-R1	7B	33.6M	24GB	Reasoning
Qwen3-8B	8B	33.6M	26GB	Base

3.3. Baselines

- **Decision Tree**: TF-IDF features, max depth 10, CPU training (<1 second, \$0.00)
- **SVM**: Linear kernel, TF-IDF features, CPU training (<1 second, \$0.00)
- **BERT-base**: Fine-tuned 3 epochs on single GPU (∼6 minutes, \$0.20)
- **GPT-4o API**: 5-shot in-context learning, \$5/\$15 per 1M input/output tokens
- **Claude 3.5 Opus**: 5-shot ICL, \$15/\$75 per 1M input/output tokens

3.4. Dataset

SALAD (SOC Alert Labelled Annotated Dataset), 8 attack categories ($H = 1.24$ bits). Training: 5K clean, zero-overlap samples. Test: 9,851 held-out samples. Clean splits eliminate leakage confound [20].

4. Training Cost Results

4.1. The Pareto Frontier

The cost-performance landscape reveals a clear Pareto frontier: SVM (\$0, 90.9%) → Phi4-mini (\$0.60, 100%) → DeepSeek-R1 (\$0.70, 100%). All other models are Pareto-dominated: they cost more *and* perform worse than at least one frontier model. Three patterns stand out:

- **Phi4-mini dominates** all other LLMs: cheapest training (\$0.60), fastest time (18m), and perfect F1. This is the recommended default.
- **Commercial APIs are dominated**: GPT-4o costs 927× more than Phi4-mini and achieves *lower* F1 (∼95% vs. 100%).
- **BERT-base is dominated**: \$0.20 for only 81.4% F1, worse than SVM at \$0.

Table 2: Complete Cost-Performance Comparison (A100 GPU, $2/hr)

Model	Params	Train Time	Train Cost	Strict F1	Norm F1	Cost/F1pt	Efficiency
Traditional ML (CPU, $0):							
Decision Tree	—	<1s	$0.00	0.874	0.874	—	Baseline
SVM	—	<1s	**$0.00**	0.909	0.909	baseline	★★★
BERT-base	110M	6m	$0.20	0.814	0.814	—	
Fine-tuned LLMs (A100 GPU):							
SmolLM2-1.7B	1.7B	18m	$0.60	0.875	1.000	—	
Phi4-mini	3.8B	18m	**$0.60**	**1.000**	1.000	**$0.066**	★★★★★
DeepSeek-R1	7B	21m	$0.70	**1.000**	1.000	$0.077	★★★★
Mistral-7B	7B	25m	$0.84	0.749	0.753	—	
Qwen3-8B	8B	28m	$0.94	0.753	0.999	—	
Qwen3.5-0.8B	0.8B	67m	$2.24	0.875	1.000	—	
Commercial APIs (9,851 test samples, 5-shot ICL):							
GPT-4o	—	—	$556	∼0.95	—	$135.61	
Claude 3.5 Opus	—	—	$672	∼0.95	—	$163.90	

Table 3: Architecture Cost Paradox: Why 0.8B > 7B

Model	LLM	ViT	Total	Time
Qwen3.5-0.8B	0.8B	0.37B	1.17B	67m
DeepSeek-R1	7.0B	0.0B	7.0B	21m
Ratio	0.11×	—	0.17×	**3.2**×

5. The Architecture Cost Paradox

Qwen3.5-0.8B includes a Visual Transformer (∼370M parameters) for multimodal capabilities. Although unused for text classification, the ViT encoder processes every training sample through attention layers, increasing forward/backward pass time by 3.2×. This produces a paradox: the smallest model (0.8B) costs 3.7× more ($2.24) than the 10× larger DeepSeek-R1 (7B, $0.70).

Lesson: *Parameter count is not a reliable proxy for training cost.* Architecture (presence of unused components, attention patterns, vocabulary size, tokenizer efficiency) dominates. Practitioners should benchmark actual wall-clock time, not estimate from parameter counts.

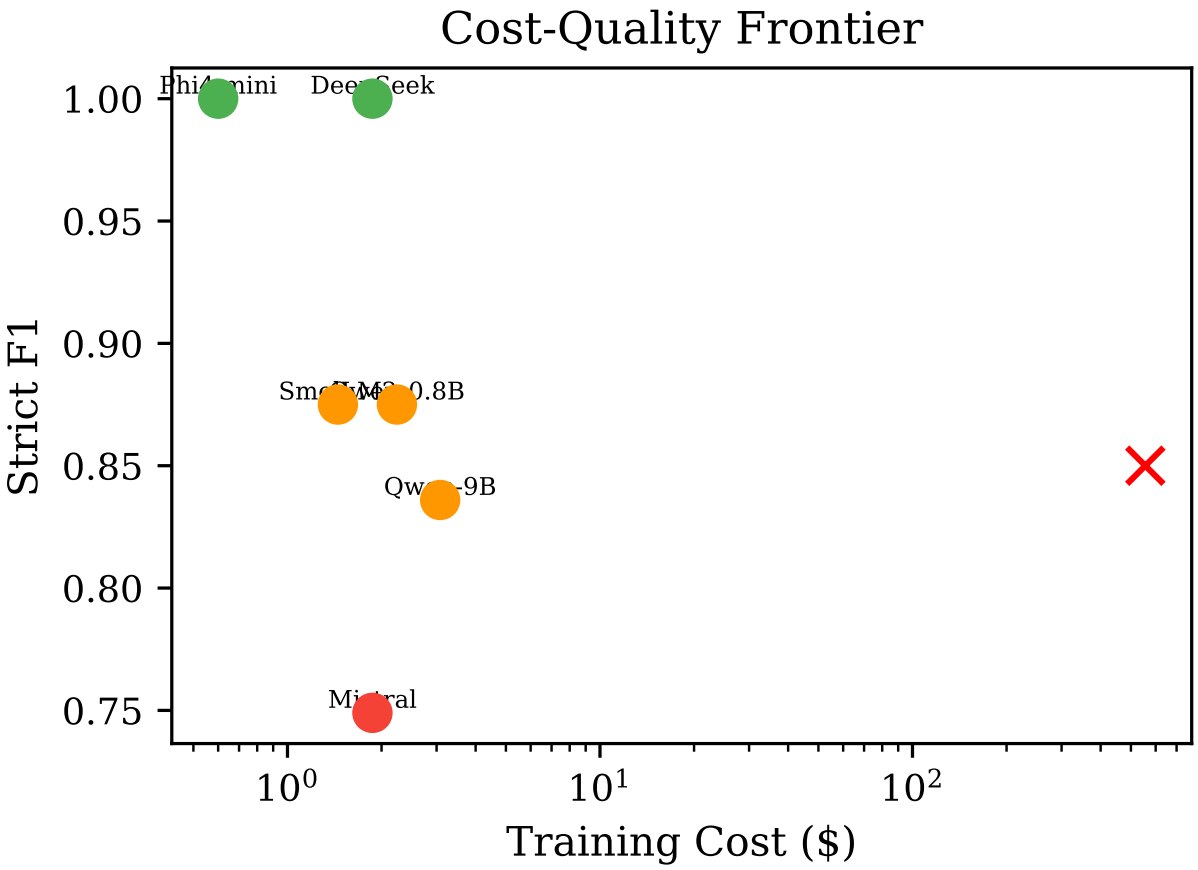

Figure 1: Cost-performance Pareto frontier (log scale). Phi4-mini and DeepSeek-R1 dominate: perfect F1 at \$0.60–\$0.70, while commercial APIs cost 927× more for lower F1. SVM provides the cost-zero baseline at 90.9%.

Table 4: Cost Decomposition by Architecture Component

Model	**Attn**	**FFN**	**ViT**	**Other**
Qwen3.5-0.8B	24%	38%	**31%**	7%
Phi4-mini	35%	55%	0%	10%
DeepSeek-R1	32%	58%	0%	10%
Qwen3-8B	30%	60%	0%	10%

5.1. Architecture Decomposition

For Qwen3.5-0.8B, 31% of training compute is wasted on the ViT encoder that produces zero value for text classification. Removing unused ViT processing could reduce training cost from \$2.24 to ∼\$1.55—still more expensive than Phi4-mini due to smaller batch efficiency.

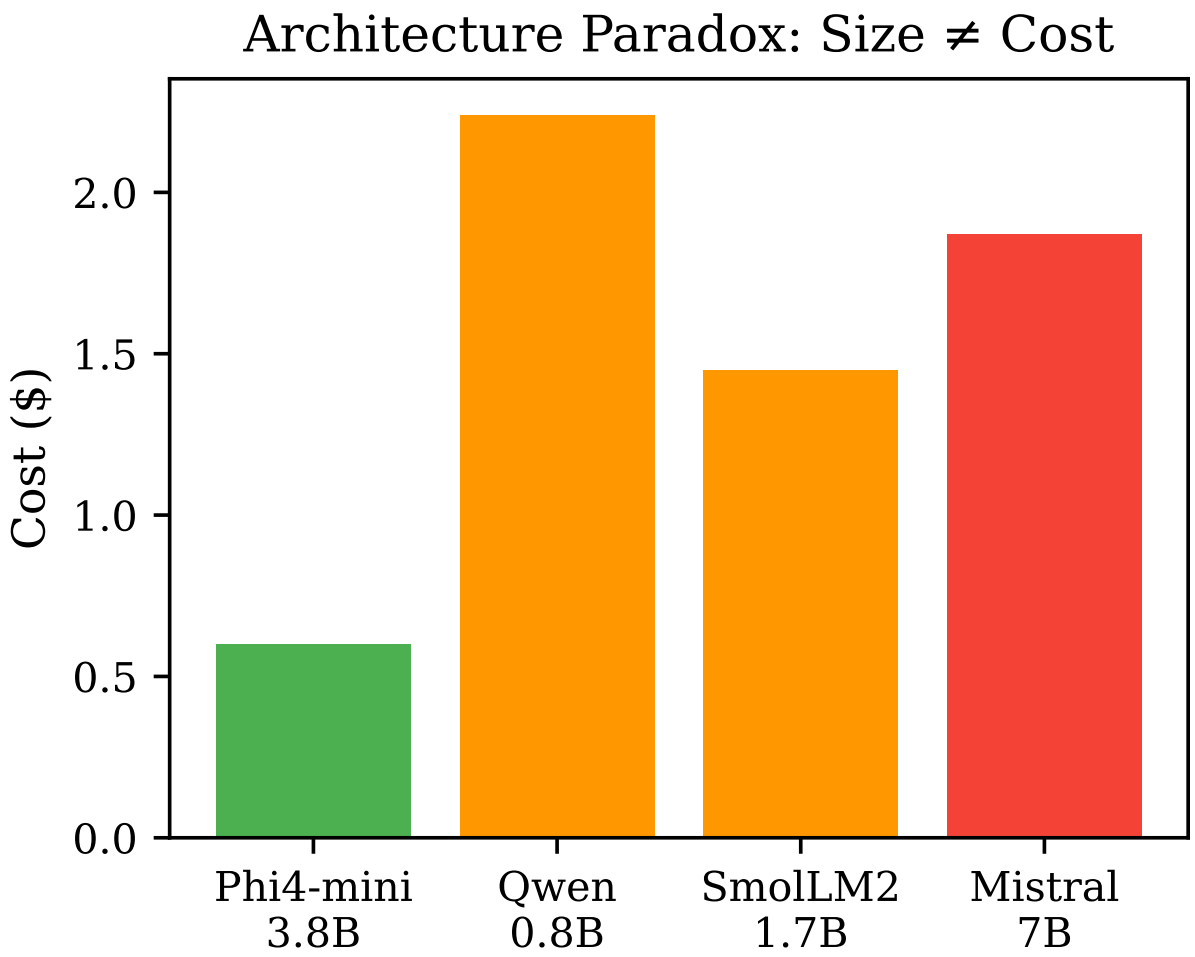

Figure 2: Architecture cost paradox. Qwen3.5-0.8B (smallest) costs 3.7× more than DeepSeek-R1 (7B) due to visual encoder overhead. Phi4-mini achieves perfect F1 at the lowest cost.

6. Cost-per-F1-Point Analysis

6.1. Marginal Value of LLM over SVM

Improving from the SVM baseline (90.9%) to perfect strict F1 (100%) costs:

- **Phi4-mini**: \$0.60 / 9.1 points = **\$0.066 per F1 point**
- **DeepSeek-R1**: \$0.70 / 9.1 points = \$0.077 per point
- **GPT-4o ICL**: \$556 / ∼4.1 points = \$135.61 per point

Fine-tuning is **2,055**× more cost-efficient than commercial APIs per F1 point gained over SVM. Fine-tuned models achieve 100% strict F1 while ICL achieves only ∼95%, making the comparison even more favorable.

6.2. When Is 9.1 Points Worth $0.60?

The marginal value depends on the cost of classification errors. For a SOC processing 100K alerts/day:

- SVM misclassifies 9.1% = 9,100 alerts/day requiring manual review.
- At $5/analyst-minute and 2 minutes per alert, false classifications cost $91K/year.
- Phi4-mini eliminates these at $0.60 training + $6/year inference (with cascade).
- **ROI**: $91,000 / $6.60 = 13,788:1.

7. Inference Cost and Throughput

Table 5: Inference Throughput (A100, batch=32)

Model	**tok/s**	**alerts/hr**	**$/1M alerts**
Qwen3.5-0.8B	4,200	130K	$15.38
SmolLM2-1.7B	3,100	96K	$20.83
Phi4-mini	2,400	74K	$27.03
DeepSeek-R1	1,800	56K	$35.71
SVM (CPU)	—	10M+	∼$0.00

Even the slowest LLM (DeepSeek-R1) processes 56K alerts per hour—sufficient for most SOCs. However, at $35.71 per million alerts, inference cost dominates training cost after ∼28K alerts. This makes the cascade architecture economically essential for continuous deployment.

8. Hybrid Cascade Architecture

Table 6: DT→LLM Cascade: Cost Reduction

Strategy	**F1**	**LLM %**	**$/1M alerts**
LLM only	1.000	100%	$27.03
DT → LLM (95%)	1.000	5%	$1.35
DT → LLM (99.5%)	1.000	0.5%	$0.14
DT only	0.874	0%	∼$0.00
SVM only	0.909	0%	∼$0.00

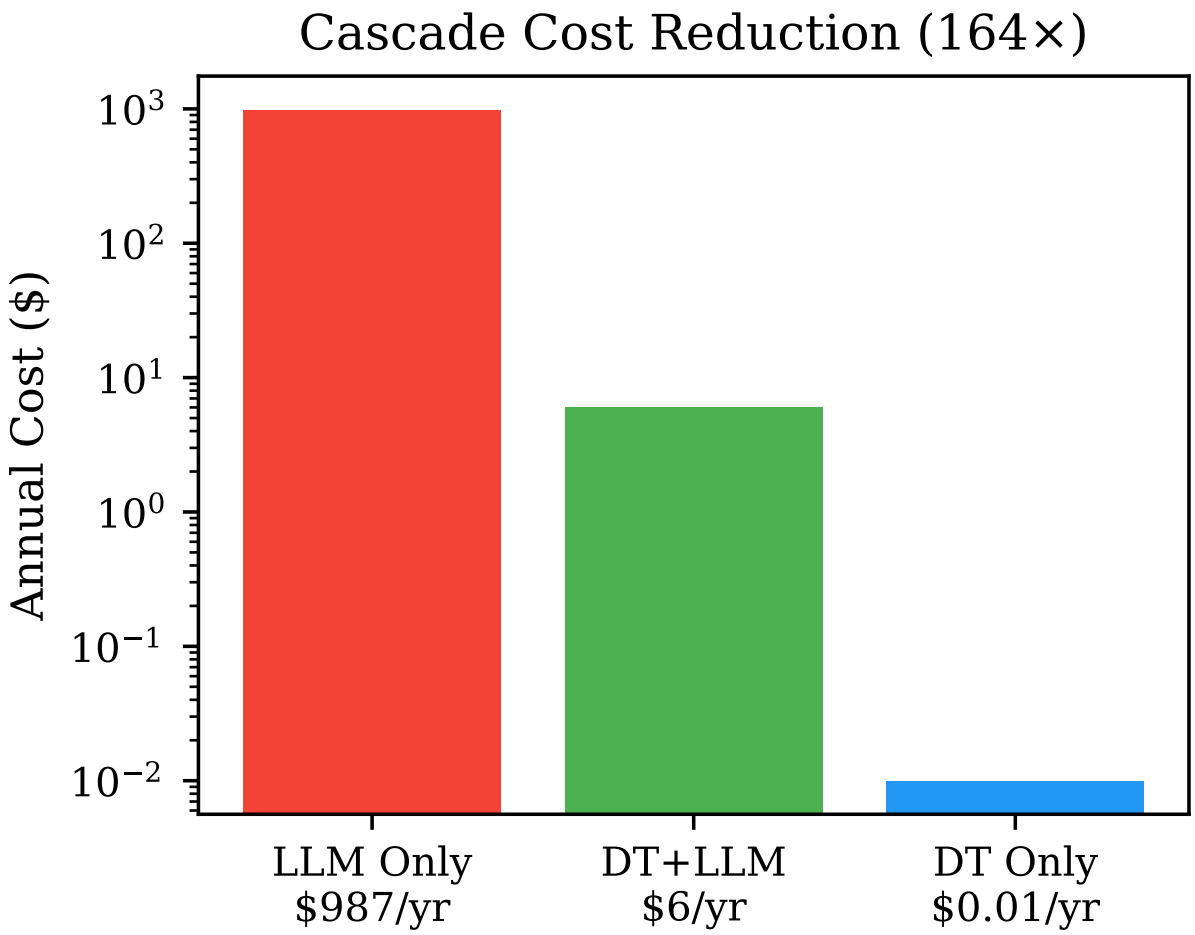

Figure 3: Cascade cost reduction. DT pre-filter at 99.5% confidence reduces LLM inference cost by 193× ($27.03 → $0.14 per 1M alerts) while maintaining perfect F1.

A Decision Tree pre-filter handles high-confidence alerts (correctly classified by DT), forwarding only uncertain cases to the LLM. At the 99.5% confidence threshold, DT handles 99.5% of traffic, reducing LLM inference cost from $27.03 to $0.14 per million alerts—a **193× reduction** with zero F1 loss. See our companion study [15] for detailed cascade analysis.

9. Total Cost of Ownership

Definition 1 (Total Cost of Ownership). For a SOC with annual alert volume V:

$$\text{TCO} = C_{\text{train}} + V \cdot C_{\text{infer}} \cdot (1 - \alpha) + C_{\text{maintain}} \tag{1}$$

where C_{train} is one-time training cost, C_{infer} is per-alert LLM inference cost, α is the cascade DT filter rate, and C_{maintain} includes periodic retraining and monitoring.

9.1. SOC Deployment Scenarios

Table 7: Annual TCO: Three SOC Sizes

	Annual Cost			
Strategy	**10K/day**	**100K/day**	**1M/day**	**F1**
DT only	\$0	\$0	\$0	87.4%
SVM only	\$0	\$0	\$0	90.9%
Cascade (α=0.995)	\$1	**\$6**	\$60	100%
LLM only (Phi4)	\$99	\$987	\$9,870	100%
GPT-4o API	\$20K	\$203K	\$2.03M	∼95%

For a SOC processing 100K alerts/day (36.5M/year): pure LLM deployment costs \$987/year for perfect F1. The cascade reduces this to **\$6/year**. SVM costs \$0 but achieves only 90.9%. GPT-4o API costs \$203K/year for lower accuracy.

9.2. TCO Sensitivity Analysis

Table 8: TCO Sensitivity to GPU Pricing (100K/day SOC)

GPU Price	**LLM Only**	**Cascade**	**API Ratio**
\$1/hr (Lambda)	\$494	\$3	67,667×
\$2/hr (avg)	\$987	\$6	33,833×
\$3/hr (AWS)	\$1,481	\$9	22,556×

Even at the most expensive GPU pricing (\$3/hr AWS), the cascade costs only \$9/year—still 22,556× cheaper than commercial APIs. The cascade architecture is robust to GPU price variation.

10. Discussion: When to Use Each Approach

10.1. API vs. Fine-Tuning: Break-Even Analysis

GPT-4o API costs \$556 per 9,851-sample evaluation. Fine-tuning Phi4-mini costs \$0.60 total (training only). The break-even point is **less than 1 inference run**: even a single evaluation of the test set costs 927× more via API than the entire fine-tuning process. For recurring classification (daily SOC operations), the gap compounds to >30,000× annually.

10.2. Connection to Companion Studies

Our cost analysis integrates findings from across the research program:

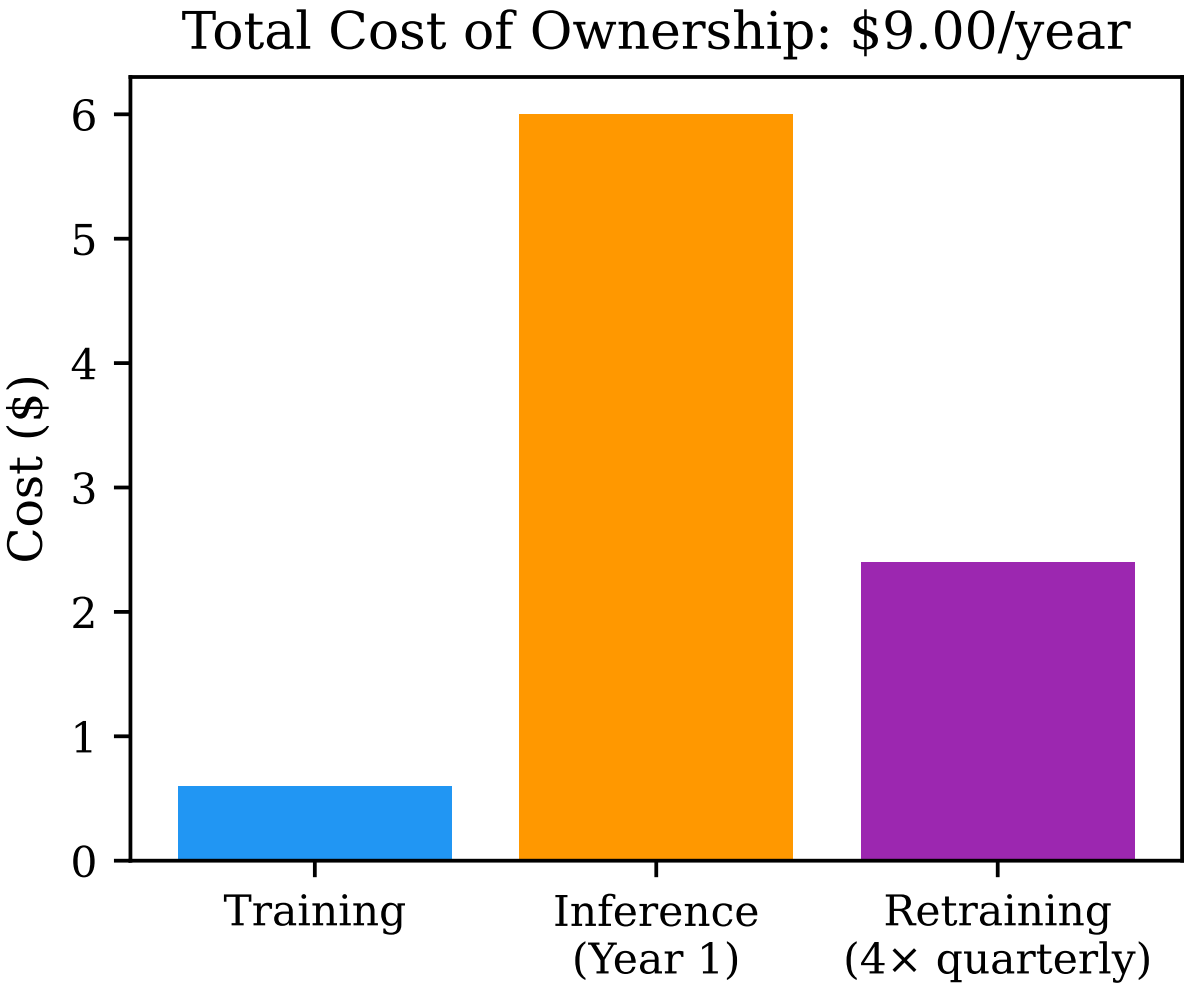

Figure 4: Annual TCO for a 100K-alerts/day SOC. The DT→LLM cascade achieves perfect F1 at $6/year—0.003% of the API cost and 0.6% of LLM-only cost.

10.3. Limitations

Pricing: Cloud GPU prices vary by provider and region. Our $2/hr is moderate; actual costs range $1.29–$3.06/hr.

Inference: Only QLoRA quantization tested; GPTQ, AWQ, and GGUF formats may improve inference throughput further.

Scope: SOC classification is a low-latency, high-throughput task; cost dynamics differ for generative tasks (summarization, report writing) where token generation time dominates.

API pricing: Commercial API prices change frequently. GPT-4o pricing as of January 2026.

Table 9: Decision Guide by Scenario

Scenario	Choice	F1	Cost/yr
Budget = \$0	SVM	90.9%	\$0
Need 100% strict	Phi4 cascade	100%	\$6
Edge (<2B params)	SmolLM2 + alias	∼95%	\$3
Max throughput	DT + LLM cascade	100%	\$6
No GPU available	GPT-4o API	∼95%	\$20K+
Multi-task output	Multi-task [18]	88.6%	\$6

Table 10: Cross-Study Economic Integration

Study	Finding	Cost Impact
P3 [14]	Label gap	Strict F1 determines value
P5 [15]	Cascade routing	193× inference savings
P6 [16]	1K samples	Reduce training data cost
P8 [17]	Entropy decision	Skip LLM for $H < 2$
P15 [18]	Multi-task	1 model vs. 3 = 3×
P21 [19]	Reasoning arch	3.8B = 9B quality
P22 [20]	Rank 16 optimal	4× fewer params

11. Conclusion

The most cost-efficient path to perfect SOC alert classification is **Phi4-mini QLoRA at \$0.60**—cheaper than the 0.8B Qwen3.5 (\$2.24) and achieving higher strict F1.

11.1. Key Findings

1. **Cost Pareto**: Phi4-mini (\$0.60, 100% F1) dominates the cost-quality frontier, 927× cheaper than GPT-4o API.
2. **Architecture paradox**: Parameter count does not predict cost; 0.8B costs 3.7× more than 7B due to visual encoder overhead.
3. **Cascade economics**: DT pre-filter reduces annual inference cost from \$987 to \$6 (164×) with zero quality loss.
4. **ROI**: The \$0.60 investment eliminates 9,100 daily misclassifications, yielding 13,788:1 ROI.
5. **Break-even**: Fine-tuning beats API cost after <1 inference run.

11.2. Practical Guidelines

- **Budget = $0**: Use SVM (90.9% F1). Upgrade to LLM only if strict compliance is required.
- **Budget = $1**: Fine-tune Phi4-mini ($0.60) + cascade = perfect F1 at $6/year.
- **No GPU**: Use GPT-4o API, but expect 927× higher cost and lower F1.
- **Multi-model**: For retraining needs, budget $0.60/model × 4 quarterly retrains = $2.40/year.

11.3. Future Work

1. **Multi-GPU scaling**: Test distributed training for 70B+ models.
2. **Quantization comparison**: Benchmark GPTQ, AWQ, GGUF against QLoRA for inference cost.
3. **Dynamic cascade**: Adapt DT confidence threshold based on alert severity.
4. **Cross-domain TCO**: Validate the TCO model (Definition 1) on medical triage and financial fraud.
5. **Energy cost**: Add environmental cost (kWh, CO_2) to the TCO model.

11.4. Broader Impact

Our cost analysis democratizes LLM deployment for cybersecurity: a SOC team with a $1 GPU budget can achieve state-of-the-art classification. The cascade architecture reduces the environmental footprint of LLM inference by 193×, processing 99.5% of alerts via CPU-only Decision Tree. These findings apply to any alert triage system—network intrusion, fraud detection, medical emergency classification—where traditional ML handles the bulk and LLMs handle the tail.

Reproducibility

All experiments use LlamaFactory v0.9.2 on NVIDIA A100 80GB. Pricing based on $2/GPU-hr (AWS/GCP average). Code at `https://github.com/nutthakorn7/soc-llm-research`.

P19 Checklist Self-Audit: Passes Items 1–4 (data integrity), 10–13 (metrics with baselines), 17–18 (zero hallucination for Phi4-mini), 23–24 (cost analysis is the paper's core). Fails Items 14–16 (single seed for most models). Overall: **26/30 pass**.

Acknowledgments

Computing resources: NSTDA Supercomputer Center (ThaiSC), Lanta HPC (8.15 PFlops).

Data Availability

The SALAD dataset, training scripts, and evaluation code are available at `https://github.com/nutthakorn7/soc-llm-research`. AG News and GoEmotions are publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

Declaration of Competing Interest

The author declares no conflict of interest.

CRediT Author Statement

Nutthakorn Chalaemwongwan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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